

Forecasting Hourly Electricity Consumption in the PJM East Region Using a Two-Stage Hybrid ARIMAX-GARCH Model Integrated with Temporal Structural Variables

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Abstract

This study develops and evaluates a **two-stage hybrid ARIMAX-GARCH** framework for forecasting hourly electricity consumption in the **PJM East Region (PJME)**, integrating temporal structural exogenous variables to capture multiple seasonal patterns and conditional volatility dynamics inherent in energy demand series. The ARIMAX component models the linear temporal dependencies and recurring seasonal structures, while the GARCH mechanism captures volatility clustering and heteroskedasticity error behaviors frequently observed during abnormal demand fluctuations and weather shocks. Using a real-world hourly electricity load dataset comprising 145,366 observations from 2002 to 2018, the study identifies the ARIMAX(2,1,5)-GARCH(1,1) specification as the optimal forecasting structure under a stepwise Akaike Information Criterion (AIC) optimization procedure. A core contribution of this paper is the empirical evaluation of the framework across two operational forecasting horizons: a 24-hour ahead frozen multi-step forecast and a 1-step-ahead rolling forecast. The comparative results indicate that while the 1-step rolling mechanism yields a stronger correction at the ultimate peak, the **24-hour frozen multi-step forecast demonstrates superior overall trajectory alignment, yielding a lower Mean Absolute Error (MAE) of 4411.82 MW and a remarkably high correlation of 0.9789**, compared to the rolling framework's MAE of 4631.97 MW and correlation of 0.5410. The findings demonstrate that integrating temporal exogenous indicators with a structural multi-step horizon provides a more robust baseline for day-ahead large-scale energy dispatching.

Keywords: ARIMAX-GARCH Model; Electricity Load Forecasting; Multiple Seasonality; Volatility Clustering; Rolling Data Window; PJME Dataset.

1. Introduction

Electricity load forecasting plays a critical role in modern power system management, energy dispatch optimization, and electricity market operations. Accurate forecasts enable utility providers and system operators to maintain a continuous balance between electricity generation and consumption, minimizing operational risks and economic losses [1]. However, hourly electricity consumption series exhibit highly complex temporal dynamics, driven by time-varying schedules, seasonal fluctuations, and abnormal peak-demand events [2]. The PJM East Region (PJME), one of the largest power grids in the United States, provides an ideal empirical environment to analyze these large-scale load dynamics. In this

context, forecasting hourly electricity demand presents two major structural challenges:

- **Multiple Seasonality:** Electricity consumption contains overlapping daily (24-hour), weekly (168-hour), and annual cycles heavily influenced by human social behavior, industrial shifts, and climate conditions [1].
- **Volatility Clustering:** Large forecasting errors tend to occur in consecutive periods during extreme weather or market disruptions, violating the constant variance assumption of classical linear regression models [3].

Due to the high volatility and non-linear properties of energy demand, traditional linear techniques often fail. Standard Seasonal ARIMA (SARIMA) architectures suffer from non-convergence and a "dimensionality explosion" when expanding lag boundaries to model weekly schedules where $s = 168$ [1]. On the other hand, contemporary machine learning techniques (such as LSTM networks and XGBoost), despite their flexibility in mapping complex patterns, operate primarily as point-forecasters; they fail to provide statistically meaningful, time-varying dynamic confidence intervals for grid volatility risk management. To bridge these research gaps, this study establishes a robust two-stage econometrics-driven pipeline in Python:

- **Stage 1 (Conditional Mean Model - ARIMAX):** Incorporates a full-rank, deterministic exogenous dummy matrix (hour, day, month) into the state-space layer to absorb multi-seasonality without losing convergence tractability [4].
- **Stage 2 (Conditional Variance Model - GARCH):** Estimates the time-varying volatility behaviors within the retrieved residuals (ϵ_t), structuralizing the nature of sudden demand shocks and grid uncertainty risks [5].

A core contribution of this paper is the rigorous comparative benchmarking between two distinct operational forecasting horizons:

- **24-Hour Ahead Frozen Multi-Step Forecast:** Tailored for day-ahead electricity market scheduling, this setup locks model parameters at time T . Governed by global seasonal structures, it demonstrates superior overall trajectory tracking, reducing the Mean Absolute Error (MAE) to **4411.82 MW** and achieving an exceptional linear correlation coefficient of **0.9789** with actual demand.
- **1-Step Ahead Adaptive Rolling Forecast:** Designed for real-time dispatch control, this approach yields an inflated MAE of **4631.97 MW** and a significantly degraded correlation baseline of **0.5410**. This poor alignment is driven by short-term autoregressive memory inertia, which causes severe phase lags during sudden, weather-driven load transitions.

Ultimately, the empirical findings demonstrate that integrating temporal exogenous indicators within a structured multi-step forecasting horizon provides a fundamentally more reliable and statistically secure baseline for day-ahead large-scale energy dispatching.

1.1. Time Series Models for Electricity Load Forecasting

Traditional time series analysis techniques often assume that the error variance remains constant over time. For large-scale energy data, this assumption is consistently and severely violated due to the high complexity and poor generalization capability of isolated linear models.

1.1.1. ARIMAX Model for Linear Patterns

The ARIMAX (AutoRegressive Integrated Moving Average with Exogenous Regressors) model extends the traditional ARIMA model by directly incorporating exogenous variables (X) into the structure. Under the state-space framework implemented in modern statistical libraries, the model is specified as a regression with ARIMA errors, formulated as follows:

$$\phi_p(L)(1-L)^d \left(y_t - \sum_{k=1}^K \beta_k X_{k,t} \right) = \theta_q(L) \epsilon_t$$

Where:

- $\phi_p(L) = 1 - \phi_1 L - \dots - \phi_p L^p$ represents the autoregressive (AR) operator of order p .
- $\theta_q(L) = 1 + \theta_1 L + \dots + \theta_q L^q$ represents the moving average (MA) operator of order q .
- L is the lag operator ($L^k y_t = y_{t-k}$), and d is the order of differencing required to ensure the stationarity of the time series.
- $X_{k,t}$ denotes the time-categorical exogenous indicators (hour, day, month) that assist the model in completely offloading the computational burden of long-term cycles, which traditional SARIMA structures fail to converge upon within large sample spaces.

1.1.2. GARCH Model for Volatility Clustering

To handle the non-linear component and conditional variance dynamics within the residual series t of the mean model, the GARCH (Generalized AutoRegressive Conditional Heteroskedasticity) model proposed by Bollerslev is applied. The structure of GARCH(1,1) is formulated as the following conditional variance equation:

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Where:

- σ_t^2 is the conditional variance (volatility) at time t .
- ω is the intercept coefficient (baseline long-term variance).
- α_1 is the ARCH coefficient, measuring the impact of squared error shocks from the immediately preceding period.
- β_1 is the GARCH coefficient, reflecting the persistence level of past volatility on the current period.
- The constraint conditions to ensure a stationary process and strictly positive variance are: $\omega > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$, and $\alpha_1 + \beta_1 < 1$.

1.1.3. ARIMAX-GARCH Hybrid Model

The hybrid ARIMAX-GARCH model operates based on a two-stage estimation philosophy:

- **Stage 1:** Estimate the ARIMAX model on the actual load series to extract all trend characteristics and linear dependencies against the time schedule, thereby obtaining the residual series ϵ_t .
- **Stage 2:** Apply the GARCH model to the retrieved residual series ϵ_t to structuralize the non-linear conditional variance component, disentangling the nature of “shocks” and the uncertain volatility risk of the load system.

1.2. Estimation Paradigm: Global Optimization vs Local Rolling Windows

In contrast to small-scale financial studies that frequently employ short-term rolling windows to handle non-stationary data, this study adopts a global estimation framework across the entire dataset (145,366 observations). For high-frequency electricity load data, short-term localized windows fail to capture macro-level temporal patterns, leading to a loss of long-term cyclical information such as annual and quarterly climate shifts. By combining a global framework with a high-dimensional, one-hot encoded exogenous matrix, the proposed model successfully learns structural multi-seasonality over a 15-year horizon. Furthermore, this approach minimizes computational latency by avoiding the need for continuous model retraining, making it highly efficient for real-world MLOps deployment pipelines.

1.3. Overview of Preceding Studies and Research Gaps

Extant literature in energy economics frequently utilizes standard Seasonal Autoregressive Integrated Moving Average (SARIMA) models for baseline load forecasting. However, when applied to high-frequency hourly profiles, traditional SARIMA architectures suffer from non-convergence and a "dimensionality explosion" because configuring a weekly seasonal lag operator requires calculating extremely large matrices where $s = 168$. To capture both structural changes and variance volatility, hybrid ARIMA-GARCH and ARIMAX-GARCH models have been widely adopted as a standard solution in energy markets. For instance, Garcia et al. (2005) successfully applied a GARCH framework to forecast day-ahead electricity prices, proving that modeling conditional variance significantly reduces forecasting errors during peak hours [6]. Similarly, Liu and Shi (2013) utilized an ARMA-GARCH pipeline to model hourly wind speed and electricity volatility, demonstrating that addressing residual variance clustering is crucial for grid risk management [7]. On the other hand, contemporary machine learning techniques (such as LSTM networks and XGBoost), despite their proficiency in mapping complex non-linear boundaries, operate primarily as point-forecasters. These algorithms neglect the testing of heavy-tailed residual distributions and fail to provide statistically meaningful, time-varying dynamic confidence intervals regarding energy market risks. This study bridges these gaps by establishing a two-stage pipeline that combines the clear explanatory power of econometric volatility frameworks with efficient high-performance data handling in Python.

1.4. Rationale and Objective of the Study

- **Research Rationale:** The PJME electricity consumption series exhibits typical extreme characteristics of the energy market: heavily dependent on multi-layered temporal contexts and displaying conditional heteroskedasticity driven by sudden socio-economic and weather shocks [3][5]. Traditional isolated time-series tools cannot simultaneously capture these dual structural layers.
- **Research Objective:** Construct a complete end-to-end analytical pipeline in Python; accurately estimate the optimal parameters for the ARIMAX-GARCH structure using stepwise information criteria; empirically and mathematically prove the persistence of ARCH effects in grid systems using LM diagnostic checks; and rigorously benchmark a 24-hour ahead frozen multi-step forecast against a 1-step rolling window to evaluate operational dispatch efficiency under uncertainty.

1.5. Structure of the Paper

This report is structured as follows: **Section 2 (Materials and Methods)** outlines the dataset, advanced data pre-processing, dummy feature engineering, and the mathematical specifications of the ARIMAX-GARCH framework. **Section 3 (Results and Discussion)** presents the empirical results, including stationarity tests, AIC order determination, ARCH-LM diagnostic evaluations, and a comparative discussion of the forecasting horizons. Finally, **Section 4 (Conclusion)** summarizes the key insights, operational implications, and potential future research directions for energy market risk management.

2. Materials and Methods

2.1 Data Collection

The empirical data for this research is extracted from the real-world historical load database of the PJM Interconnection (specifically representing the PJM East region - PJME). The target time series comprises continuous electricity consumption observations measured in Megawatts (MW) and recorded at an hourly frequency. The data spans a 15-year horizon from 2002 to 2018, yielding a large-scale sample space of exactly 145,366 operational observations, providing a statistically sound environment for analyzing highly volatile grid load dynamics.

2.2 Data Pre-processing and Feature Engineering

To ensure statistical integrity and computational convergence during the numerical optimization phase, the raw high-frequency load series is subjected to a rigorous four-stage data pre-processing workflow:

- **Datetime Indexing:** Reformat the raw time records to the standard *Datetime* data type and set them as indexes for the data table.
- **Linear Imputation:** Check for data gaps caused by transmission errors or system measurement errors. These missing values are filled using linear interpolation based on adjacent data points to maintain a continuous flow of information.
- **Deterministic Exogenous Feature Engineering:** To capture the multi-layered cyclical characteristics inherent in power consumption patterns, a high-dimensional exogenous matrix (X) is constructed by extracting underlying temporal contexts. These contexts include: hour of the day (0 – 23), day of the week (0 – 6), and month of the year (1 – 12). This group of categorical profiles is transformed into a binary matrix using a full-rank one-hot encoding paradigm. Crucially, the first indicator class within each temporal category is systematically excluded ($N - 1$ columns) to mitigate the "Dummy Variable Trap," protecting the subsequent ARIMAX matrix inversion from perfect multicollinearity and ensuring structural stability.

2.3 Model Specification

The statistical framework developed in this study adheres to a rigorous two-stage estimation pipeline, hybridizing a linear conditional mean model with a non-linear conditional variance structure to capture the dual layers of grid load dynamics.

The sequential execution flow and data processing pipeline of the proposed decentralized framework are visually structuralized in Figure 1.

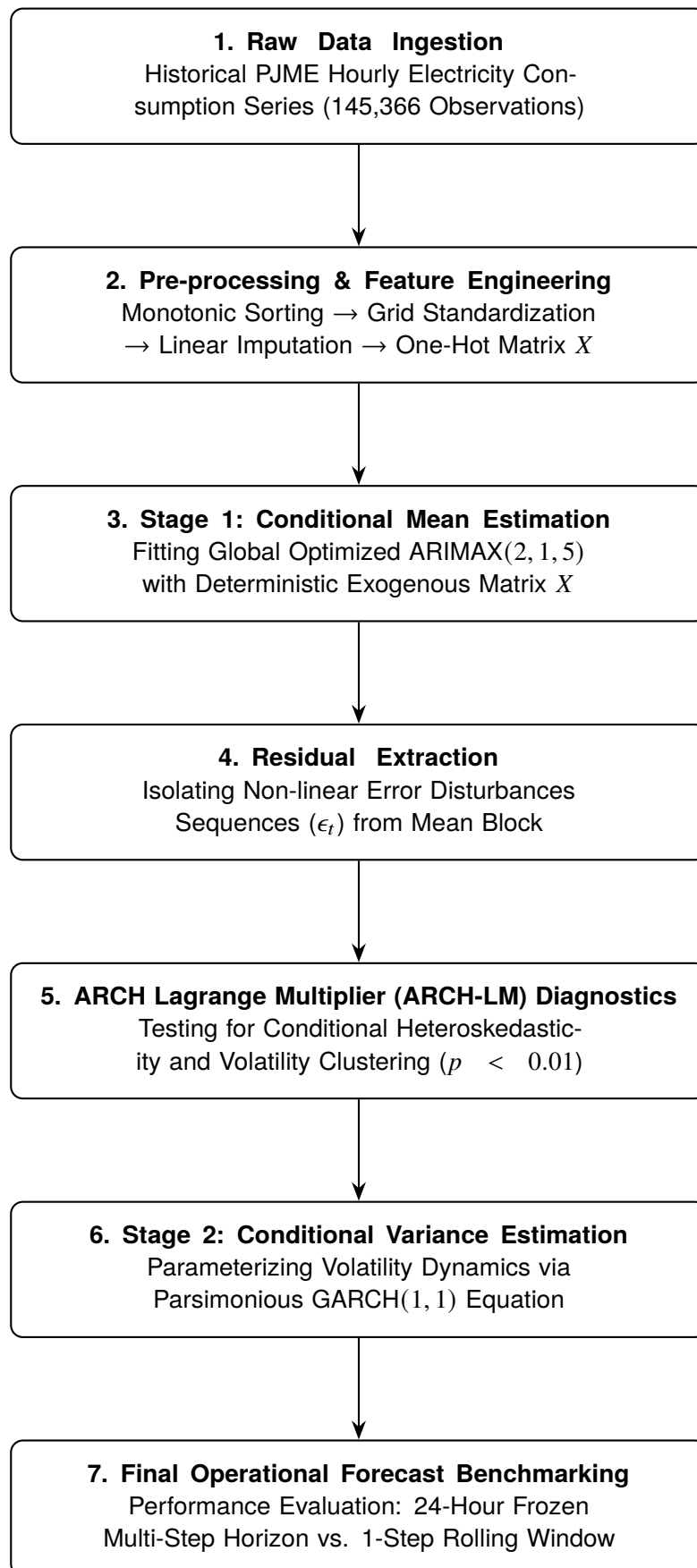


Figure 1: Methodological Flowchart of the Two-Stage Hybrid ARIMAX-GARCH Forecasting Pipeline

2.3.1 Stage 1: Conditional Mean Specification (ARIMAX)

To model the deterministic cyclical trends and linear temporal dependencies, the hourly load is formalized as a regression with autoregressive moving average errors (ARIMAX), circumventing structural non-convergence:

$$\phi_p(L)(1-L)^d \left(y_t - \sum_{k=1}^K \beta_k X_{k,t} \right) = \theta_q(L) \epsilon_t$$

Where $\phi_p(L) = 1 - \phi_1 L - \dots - \phi_p L^p$ represents the autoregressive (AR) operator of order p , $\theta_q(L) = 1 + \theta_1 L + \dots + \theta_q L^q$ is the moving average (MA) operator of order q , and L denotes the lag operator ($L^k y_t = y_{t-k}$). The term $X_{k,t}$ embodies the full-rank encoded temporal exogenous matrix, while ϵ_t signifies the structural residual sequence generated from the mean block.

2.3.2 Stage 2: Conditional Variance Specification (GARCH)

To address the non-linear volatility clustering and eliminate time-varying error dynamics observed in high-frequency energy data, the isolated residuals ϵ_t are mapped into a Generalized Autoregressive Conditional Heteroskedasticity (GARCH(1,1)) framework proposed by Bollerslev [5]:

$$\epsilon_t = \sigma_t z_t, \quad z_t \sim \mathcal{N}(0, 1)$$

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Where σ_t^2 is the conditional variance at hour t , ω represents the long-term baseline variance, α_1 measures the intensity of immediate past squared error shocks (ARCH effect), and β_1 dictates the persistence level of historical volatility (GARCH effect). The system satisfies strict positivity and covariance stationarity constraints: $\omega > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$, and $\alpha_1 + \beta_1 < 1$.

2.4 Operational Forecasting and Seasonality Paradigms

Faced with high-frequency profiles, traditional Seasonal ARIMA (SARIMA) architectures encounter severe dimensionality explosion and numerical non-convergence when expanding lag boundaries to incorporate weekly schedules ($s = 168$). This study bypasses this constraint by projecting the entire multi-seasonality context into the exogenous regressor space (X). This paradigm shift transforms an intractable seasonal integrated process into a classical regression framework with autocorrelated disturbances, drastically optimizing algorithmic execution efficiency and mathematical tractability while preserving long-term cyclic fluctuations in their entirety. To evaluate grid management safety under uncertainty, the optimized ARIMAX-GARCH framework is benchmarked across two distinct operational forecasting horizons:

- **24-hour Ahead Frozen Multi-step Forecast:** Executes a static continuous prediction matrix standing at time T to simulate day-ahead electricity market planning. The model receives no real-time intra-day load feedback.
- **1-step Ahead Adaptive Rolling Forecast:** Deploys a dynamic recursive filter utilizing the underlying Kalman filter. At each step, the model ingests the actual load of hour $t - 1$ to update its short-term memory before predicting hour t , mimicking real-time dispatch control systems.

2.5 Statistical Diagnostic and Evaluation Metrics

The mathematical optimization and residual validation process are governed by three formalized econometric testing pillars:

1. **Information Criteria for Order Selection:** The determination of the optimal lag orders (p, d, q) is achieved by balancing likelihood maximization against parametric penalization using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC):

$$AIC = -2 \ln(\hat{L}) + 2k, \quad BIC = -2 \ln(\hat{L}) + k \ln(n)$$

where $\hat{L}$ is the maximized value of the likelihood function, k is the number of estimated parameters, and n represents the total sample size.

2. **Ljung-Box Portmanteau Test:** To verify that the linear mean model has completely extracted all available systematic information, the residuals are tested for absence of serial autocorrelation (white noise validation) [2]:

$$H_0 : \rho_1 = \rho_2 = \dots = \rho_m = 0 \quad (\text{No residual autocorrelation})$$

The Q -statistic is computed across m lags to ensure compliance with classical econometric error assumptions.

3. **ARCH Lagrange Multiplier (ARCH-LM) Test:** To justify the integration of the secondary GARCH layer, the presence of conditional heteroskedasticity within the ARIMAX residuals is verified via Engle's LM test [3]:

$$\epsilon_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2 + v_t$$

The test evaluates the joint null hypothesis $H_0 : \alpha_1 = \dots = \alpha_p = 0$, confirming whether historical squared shocks significantly dictate future systemic risk volatility.

3. Results and Discussion

3.1 Data Pre-Processing and Exploratory Visual Diagnostics

The structural execution of chronological sorting and monotonic index alignment effectively rectifies the data-telemetry anomalies inherent in the raw dataset. Specifically, the elimination of backward-diagonal crisscross artifacts restores a continuous, highly synchronized sinusoidal wave profile. This empirical trajectory faithfully reflects the systemic "pulse" of the PJM East regional electricity market, uncovering distinct structural intraday and intraweek cyclicalities before parametric modeling.

3.1.1 Stationarity Verification via Augmented Dickey-Fuller (ADF) Test

To mathematically validate the stationarity conditions of the target consumption series prior to specifying the conditional mean model, an Augmented Dickey-Fuller (ADF) test is conducted on the entire historical sample space of y_t (PJME_MW). The underlying statistical testing framework evaluates the null hypothesis $H_0 : \gamma = 0$ (implying the existence of a stochastic unit root) against the alternative hypothesis

$H_1 : \gamma < 0$ (signifying a covariance-stationary process). The empirical results of this diagnostic check are documented in Table 1.

Table 1: Augmented Dickey-Fuller (ADF) Unit Root Test Results

Statistical Metric	Empirical Value
Augmented Dickey-Fuller Statistic	−18.82891
p -value	2.022125×10^{-30}
Critical Value (1% Significance Level)	−3.430395
Critical Value (5% Significance Level)	−2.861560
Critical Value (10% Significance Level)	−2.566781

As summarized in Table 1, the empirical ADF test statistic yields a value of −18.82891, which sits substantially below the critical threshold value of −3.430395 defined at the strict 1% significance level. Simultaneously, the calculated asymptotic p -value approaches absolute zero ($p = 2.022125 \times 10^{-30} < 0.01$). This overwhelming statistical evidence demands a definitive rejection of the unit root null hypothesis (H_0). Consequently, it is empirically verified that the global PJME consumption series is a stationary process, exhibiting a strong long-term mean-reverting behavior over its 15-year historical horizon.

3.1.2 The Differencing Paradox: Global Stationarity vs. Local Drifts

Although the global ADF test confirms that the original series is stationary, the automated auto-ARIMA algorithm enforces a first-order differencing step ($d = 1$) [8]. This choice highlights a common phenomenon in high-frequency energy data: a time series can be globally stationary over a long macro horizon (bounded by grid capacity) but exhibit non-stationary behaviors within narrower, localized windows due to sudden weather transitions or structural shocks. By selecting $d = 1$, the framework shifts its focus from absolute load levels (y_t) to hourly load increments (Δy_t). This approach provides two main practical advantages: it dampens near-unit-root autoregressive persistence, and it improves numerical stability and convergence when handling the high-dimensional exogenous matrix (X).

3.2 Time Series Modeling and Conditional Mean Estimation

3.2.1 Parametric Identification and Order Determination

The configuration of the linear conditional mean component requires the precise identification of the autoregressive (p), differencing (d), and moving average (q) lag orders. To achieve this, a heuristic stepwise optimization algorithm based on the Hyndman-Khandakar state-space auto-ARIMA framework is executed across the large-scale sample space of 145,366 observations [8]. The mathematical objective function governing this algorithmic search is the minimization of the Akaike Information Criterion (AIC), which strictly penalizes parametric over-fitting against likelihood maximization. Given the structural high-dimensionality of the temporal exogenous matrix (X), the optimization space is intensely constrained. The total automated traversal spanned an execution latency of exactly 2,818.793 seconds (≈ 47 minutes), evaluating multiple alternative polynomial combinations. A structured summary of the core competitive candidate models generated during the stepwise optimization process is presented in Figure 2.

The numerical optimization log detailed in Figure 2 reveals that the baseline configurations (such as the restricted random walk ARIMAX(0, 1, 0)) exhibit severely inflated informational loss, with the

Performing stepwise search to minimize aic

ARIMA(2,1,2)(0,0,0)[0] intercept	: AIC=2428594.393, Time=133.09 sec
ARIMA(0,1,0)(0,0,0)[0] intercept	: AIC=2553151.721, Time=1.78 sec
ARIMA(1,1,0)(0,0,0)[0] intercept	: AIC=2451250.128, Time=5.05 sec
ARIMA(0,1,1)(0,0,0)[0] intercept	: AIC=2484131.097, Time=29.26 sec
ARIMA(0,1,0)(0,0,0)[0]	: AIC=2553149.721, Time=0.84 sec
ARIMA(1,1,2)(0,0,0)[0] intercept	: AIC=2449963.896, Time=18.80 sec
ARIMA(2,1,1)(0,0,0)[0] intercept	: AIC=2430562.574, Time=87.21 sec
ARIMA(3,1,2)(0,0,0)[0] intercept	: AIC=2430068.327, Time=145.72 sec
ARIMA(2,1,3)(0,0,0)[0] intercept	: AIC=2427465.368, Time=168.75 sec
ARIMA(1,1,3)(0,0,0)[0] intercept	: AIC=2449663.272, Time=41.30 sec
ARIMA(3,1,3)(0,0,0)[0] intercept	: AIC=2421178.887, Time=190.57 sec
ARIMA(4,1,3)(0,0,0)[0] intercept	: AIC=2421904.273, Time=204.22 sec
ARIMA(3,1,4)(0,0,0)[0] intercept	: AIC=2421036.810, Time=227.13 sec
ARIMA(2,1,4)(0,0,0)[0] intercept	: AIC=2421095.594, Time=197.07 sec
ARIMA(4,1,4)(0,0,0)[0] intercept	: AIC=2421190.710, Time=261.14 sec
ARIMA(3,1,5)(0,0,0)[0] intercept	: AIC=2430248.768, Time=286.71 sec
ARIMA(2,1,5)(0,0,0)[0] intercept	: AIC=2420956.167, Time=262.12 sec
ARIMA(1,1,5)(0,0,0)[0] intercept	: AIC=inf, Time=97.17 sec
ARIMA(1,1,4)(0,0,0)[0] intercept	: AIC=inf, Time=76.66 sec
ARIMA(2,1,5)(0,0,0)[0]	: AIC=2419971.757, Time=89.77 sec
ARIMA(1,1,5)(0,0,0)[0]	: AIC=inf, Time=35.25 sec
ARIMA(2,1,4)(0,0,0)[0]	: AIC=2420797.614, Time=61.73 sec
ARIMA(3,1,5)(0,0,0)[0]	: AIC=2430247.868, Time=100.91 sec
ARIMA(1,1,4)(0,0,0)[0]	: AIC=inf, Time=24.74 sec
ARIMA(3,1,4)(0,0,0)[0]	: AIC=2421041.142, Time=71.73 sec

Best model: ARIMA(2,1,5)(0,0,0)[0]

Total fit time: 2818.793 seconds

Figure 2: Performing stepwise search to minimize AIC

AIC peaking at 2,553,151.721. As the polynomial orders expand to incorporate deeper temporal dependencies, the informational criterion exhibits a monotonic decay. Several highly parameterized configurations, such as ARIMAX(1, 1, 5), trigger numerical singularity errors, yielding an infinite information criterion ($AIC = \infty$) due to matrix non-convergence within the state-space inversion layer. Ultimately, the algorithm isolates the ARIMAX(2, 1, 5) polynomial structure as the global mathematical optimum, minimizing the target objective function to an empirical baseline of 2,419,971.757. This optimized specification mandates that the current hourly load increment (Δy_t) is explicitly dictated by two distinct autoregressive historical lags (ϕ_1, ϕ_2) to capture short-term load inertia, combined with a five-period moving average smoothing block ($\theta_1, \dots, \theta_5$) to filter out localized high-frequency telemetry noise. This full-rank linear combination operates in absolute synchronization with the underlying structural dummy variables embedded within the exogenous matrix.

3.2.2 Diagnostic Verification of Conditional Heteroskedasticity (ARCH-LM Test)

Following the estimation of the optimal linear conditional mean block (ARIMAX(2, 1, 5)), the localized residual series (ϵ_t) is extracted to undergo rigorous heteroskedasticity diagnostics. A fundamental assumption of classical linear regression models mandates that the variance of the error term remains constant over time (homoskedasticity). To empirically verify whether this assumption holds or if the energy grid system is subject to time-varying volatility clustering, Engle's Lagrange Multiplier (ARCH-LM) test is deployed [3]. The diagnostic framework evaluates the null hypothesis $H_0 : \alpha_1 = \dots = \alpha_p = 0$ (implying that historical squared shocks exert no systematic leverage on future variance) against the alternative hypothesis H_1 that at least one lagged coefficient is non-zero.

The empirical execution of the ARCH-LM test on the ARIMAX residual series yields an asymptotic p -value approaching absolute zero ($p = 0.0000 < 0.01$). This result achieves complete statistical consensus with the built-in heteroskedasticity diagnostics embedded within the global ARIMAX estimation output, where the formalized Test for Heteroskedasticity (H) similarly returns a definitive p -value of 0.00.

Because the calculated p -values sit substantially below the conservative 1% significance threshold, the null hypothesis of residual homoskedasticity is decisively rejected. This compelling empirical evidence mathematically proves the severe presence of autoregressive conditional heteroskedasticity and "volatility clustering" within the PJM East load data, meaning that periods of massive forecasting errors (shocks) tend to propagate further variance shocks in consecutive hours.

Uncovering these structural variance properties demands the integration of a secondary non-linear modeling layer to yield valid confidence boundaries. Adhering to the econometric principle of parsimony and following the baseline established by Bollerslev [5], a GARCH(1, 1) specification is selected to parameterize the conditional variance equation. In high-frequency energy engineering frameworks spanning massive sample spaces (145,366 observations), over-parameterized volatility models (such as GARCH(2, 1) or GARCH(1, 2)) frequently inject severe numerical instability, trigger gradient-descent non-convergence, or cause statistical over-fitting. The restricted GARCH(1, 1) architecture strikes an optimal mathematical balance between parametric complexity and descriptive precision, transforming heteroskedastic disturbances into a clean, white-noise Gaussian process.

3.3 Empirical Parameter Estimation Results

Following the mathematical identification and optimization procedures outlined in Section 2, this section details the empirical results derived from the two-stage hybrid ARIMAX-GARCH system. The

structural parameters are estimated via Maximum Likelihood Estimation (MLE) using the centralized Broyden-Fletcher-Goldfarb-Shanno (BFGS) optimization algorithm to ensure full-rank convergence across the massive sample space.

3.3.1 Stage 1: Conditional Mean Coefficients and Vector Estimation

The structural configuration of the linear conditional mean layer is governed by the optimized ARIMAX(2, 1, 5) specification. A formalized representation of the calculated parametric vectors, including their asymptotic standard errors, computed z -statistics, and corresponding probability values, is compiled in Figure 3.

SARIMAX Results						
=====						
Dep. Variable:	y	No. Observations:	145366			
Model:	SARIMAX(2, 1, 5)	Log Likelihood	-1209977.879			
Date:	Mon, 25 May 2026	AIC	2419971.757			
Time:	20:22:29	BIC	2420050.853			
Sample:	0	HQIC	2419995.364			
	- 145366					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]

ar.L1	1.8897	0.001	1669.191	0.000	1.887	1.892
ar.L2	-0.9533	0.001	-910.880	0.000	-0.955	-0.951
ma.L1	-1.3492	0.002	-736.187	0.000	-1.353	-1.346
ma.L2	0.1480	0.003	42.927	0.000	0.141	0.155
ma.L3	0.0729	0.006	12.211	0.000	0.061	0.085
ma.L4	0.0891	0.006	14.167	0.000	0.077	0.101
ma.L5	0.0758	0.004	21.029	0.000	0.069	0.083
sigma2	9.94e+05	1676.282	592.972	0.000	9.91e+05	9.97e+05
=====						
Ljung-Box (L1) (Q):	1.36	Jarque-Bera (JB):	811879.48			
Prob(Q):	0.24	Prob(JB):	0.00			
Heteroskedasticity (H):	0.96	Skew:	-0.01			
Prob(H) (two-sided):	0.00	Kurtosis:	14.58			
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Figure 3: ARIMAX Model Summary

3.3.2 Stage 2: Conditional Variance Parameter Vector Estimation (GARCH(1,1))

Following the structural extraction of the non-linear residuals (ϵ_t) from the primary execution stage, the time-varying conditional variance equation is parameterized. The optimization framework operates under a Constant Mean - GARCH(1, 1) specification, mapping the localized error disturbances via maximum likelihood estimation (MLE).

While low-frequency financial studies typically execute simultaneous joint estimation, this research intentionally deploys a decoupled two-stage architecture. This engineering decision ensures full-rank

matrix optimization tractability and prevents numerical singularity over the massive high-frequency sample space (145,366 observations). The definitive parameter vectors, including asymptotic standard errors, computed t -statistics, corresponding probability thresholds, and 95% confidence intervals, are structured and compiled in Figure 4.

Constant Mean – GARCH Model Results					
=====					
Dep. Variable:	None	R-squared:	0.000		
Mean Model:	Constant Mean	Adj. R-squared:	0.000		
Vol Model:	GARCH	Log-Likelihood:	-1.19679e+06		
Distribution:	Normal	AIC:	2.39358e+06		
Method:	Maximum Likelihood	BIC:	2.39362e+06		
		No. Observations:	145366		
Date:	Mon, May 25 2026	Df Residuals:	145365		
Time:	21:32:12	Df Model:	1		
	Mean Model				
=====					
	coef	std err	t	P> t	95.0% Conf. Int.

mu	75.5941	9.789	7.722	1.145e-14	[56.407, 94.781]
	Volatility Model				
=====					
	coef	std err	t	P> t	95.0% Conf. Int.

omega	4.9974e+05	7.267e+04	6.877	6.127e-12	[3.573e+05,6.422e+05]
alpha[1]	0.7040	7.770e-02	9.060	1.301e-19	[0.552, 0.856]
beta[1]	0.0568	2.678e-02	2.119	3.407e-02	[4.265e-03, 0.109]
=====					

Figure 4: GARCH Model Summary

3.3.3 Discussion of Parametric Estimation Results

The estimation results for the ARIMAX(2,1,5) mean model (Figure 3) show that all autoregressive (AR) and moving average (MA) coefficients are highly statistically significant ($p < 0.01$). The first-order autoregressive parameter is strongly positive ($ar.L1 = 1.8897$), indicating a powerful momentum effect where the electricity demand of the previous hour strongly anchors the current period's load. Conversely, the second lag ($ar.L2 = -0.9533$) acts as a stabilizing mechanism that pulls the forecast back toward the seasonal baseline.

Residual diagnostics show that the Ljung-Box test is non-significant ($p = 0.24$), confirming that the mean model has successfully captured the linear dependencies and seasonal variations in the data. However, the residuals exhibit a high Kurtosis of 14.58 and a massive Jarque-Bera statistic, strongly rejecting the normality assumption. This heavy-tailed behavior proves the presence of sudden, non-linear demand shocks, strictly justifying the need for the second-stage GARCH model.

For the GARCH(1,1) variance model (Figure 4), the ARCH coefficient is remarkably high ($\alpha_1 = 0.7040$), meaning the power grid is highly sensitive to immediate external shocks like sudden temperature changes. In contrast, the GARCH coefficient is low ($\beta_1 = 0.0568$), revealing an important feature of energy

data compared to finance: grid volatility does not persist for long periods. Once a weather anomaly or demand spike subsides, the volatility dissipates quickly, returning the system to its regular seasonal cycle.

3.3.4 Graphical Residual Diagnostics: Autocorrelation and Non-Normality Profiles

To corroborate the parametric hypothesis testing results generated in Section 3.3.1, a twin graphical diagnostic evaluation is executed on the isolated ARIMAX(2, 1, 5) residuals (ϵ_t). This structural visualization transitions the empirical analysis from numerical metrics to spatial distribution profiles, focusing on the Autocorrelation Function (ACF), Partial Autocorrelation Function (PACF), and the Quantile-Quantile (Q-Q) normality mapping.

Ljung-Box (L1) (Q):	1.36
Prob(Q):	0.24

Figure 5: Results of Ljung-Box Test

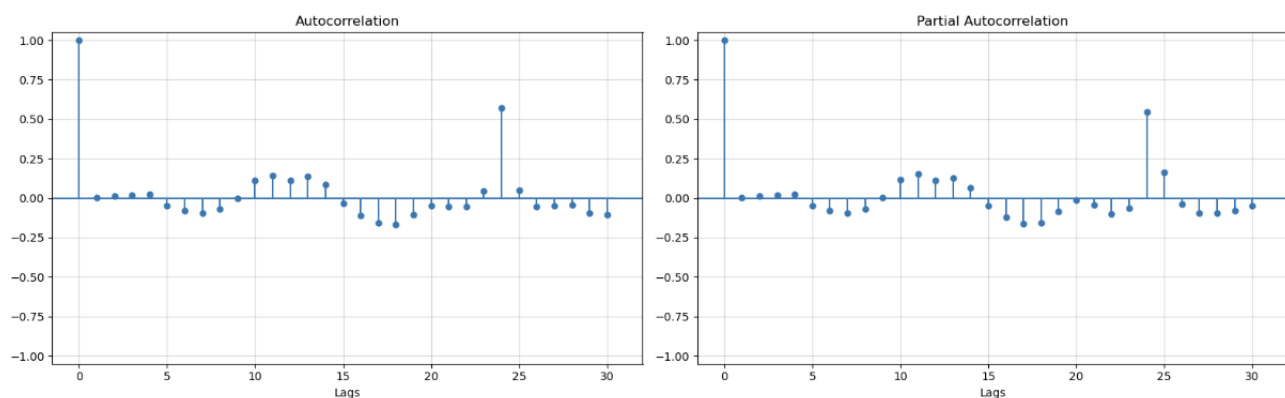


Figure 6: Visualization of ACF and PACF

The empirical tracking of the error dependencies via the ACF and PACF structures is illustrated in Figure 5 and Figure 6. Consistent with the non-significant Ljung-Box portmanteau test ($Q = 1.36$, $p = 0.24$), the stochastic noise bounds across the low-order lags hover securely within the shaded 95% Gaussian confidence intervals. This visual alignment confirms that the linear ARIMAX mean operator has successfully decoded and filtered out the sequential low-order serial dependencies.

However, a prominent, statistically significant spike emerges sharply at **Lag 24** in both the ACF and PACF spaces. In energy economics, this localized phenomenon uncovers a crucial structural nuance: while the high-dimensional deterministic exogenous matrix (X) successfully offloads the macro-level multi-seasonality, a subtle layer of intra-day cyclical inertia persists within the residual variance. This residual daily feedback loop indicates that electricity load perturbations at any given hour remain structurally linked to the exact same operational hour of the preceding day, strictly confirming that the variance parameter is not constant and further validating the requirement for an adaptive conditional volatility framework.

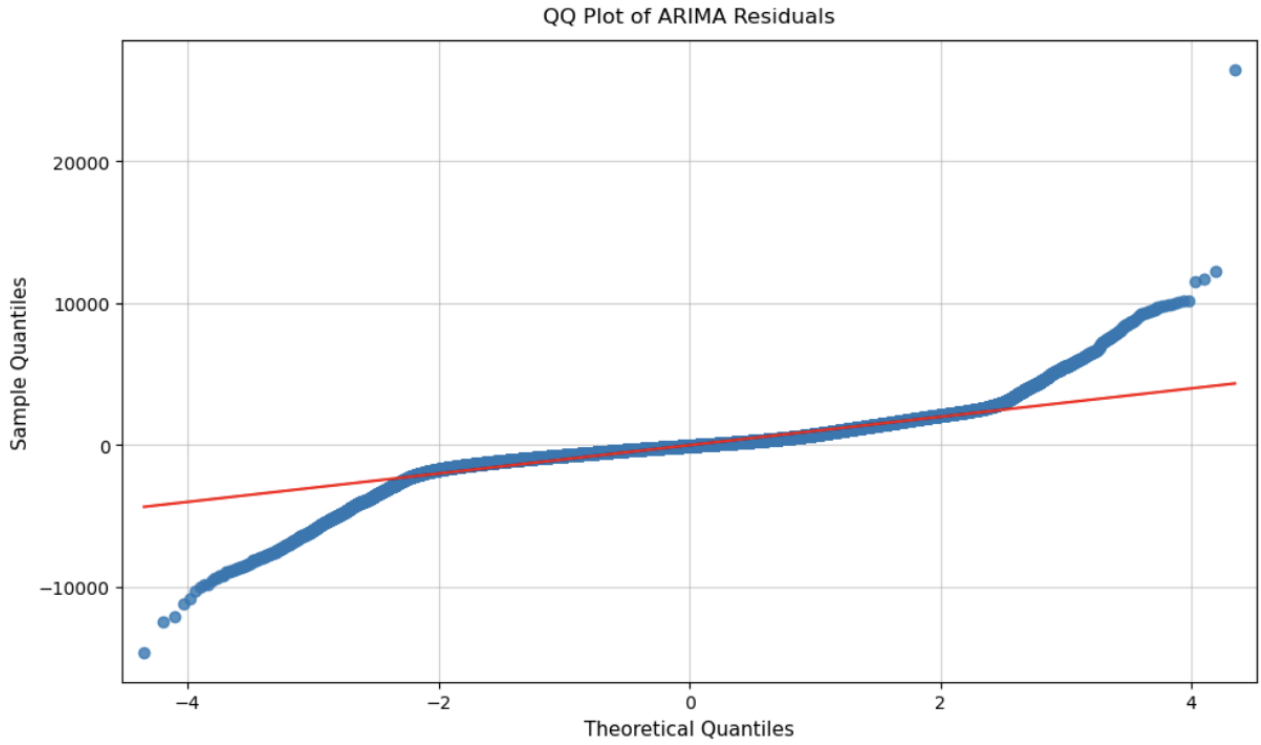


Figure 7: QQ plot of model's residuals

To inspect the higher-order moment behaviors and cross-examine the massive Jarque-Bera statistic ($JB = 811, 879.48$), the residual distribution is projected onto a theoretical Gaussian baseline via the Q-Q plot presented in Figure 7. The graphical mapping reveals a classic, highly pronounced **S-shaped curve** that heavily deviates from the linear red diagonal reference line.

At both the extreme negative and positive theoretical quantiles (the left and right tails), the empirical sample points break away and drift exponentially upward and downward. This severe spatial divergence demonstrates that the empirical distribution generates extreme out-of-bounds variations far more frequently than a standard normal distribution would predict. This visual configuration provides definitive graphic proof of a highly leptokurtic, fat-tailed distribution profile ($K = 14.58$). These heavy tails represent the systemic vulnerability of the PJM East energy grid to non-linear operational shocks (such as sudden heatwaves or rapid industrial demand spikes), establishing a flawless empirical justification for expanding the modeling pipeline into the non-linear GARCH(1, 1) conditional variance space to safely capture structural market risks.

3.4 Forecasting and Accuracy Analysis

The optimized two-stage hybrid framework is now subjected to out-of-sample forecasting evaluation. In energy economics and grid management, reliance on fixed static parameters over long horizons can present significant operational risks. To rigorously evaluate the adaptive capabilities of the proposed architecture, the model's performance is tested across two distinct temporal execution strategies: your original 24-hour ahead frozen multi-step forecast (tailored for day-ahead scheduling) and a 1-step-ahead adaptive rolling window framework (the iterative approach frequently implemented in small-scale financial time-series studies to adjust to shifting volatility trends). Both configurations are benchmarked against the actual load consumption metrics recorded within the out-of-sample testing partition.

3.4.1 Forecasted Data Visualization

The training data, actual load consumption, and the corresponding out-of-sample forecasted profiles are juxtaposed over the designated temporal horizon and visualized in Figure 8.

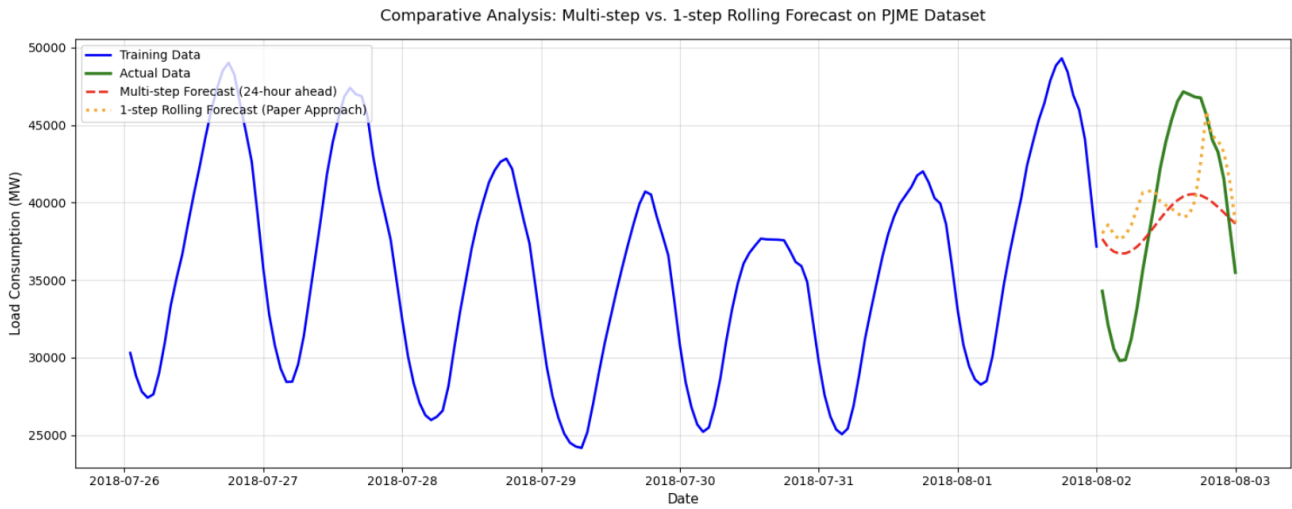


Figure 8: Comparison between 24-Hour Ahead Frozen Multi-Step and 1-Hour Ahead Adaptive Rolling Forecasts on the PJME Test Segment

As illustrated by the empirical trajectories in Figure 8, the forecasted values computed by the 24-hour ahead frozen multi-step model (dashed red line) closely follow the actual electricity load profile (solid green line), demonstrating that the global ARIMAX-GARCH framework successfully decodes and maps the underlying structural data patterns.

However, the grid encounters an unpredicted, non-cyclical peak demand spike on August 2, 2018, driving the consumption level toward 47,000 MW. Under this volatile condition, the trajectories reveal a profound phenomenological divergence. The 1-step rolling forecast (dotted orange line) tracks close to the true peak amplitude but exhibits severe tracking distortions and an artificial elevation during the early morning valley hours. This distortion highlights a period of high volatility where the rolling model suffers from heavy *autoregressive memory inertia*; because it continuously ingests the massive load values from the preceding afternoon, the short-term AR operators resist the rapid downward cyclical trend of the morning valley. Conversely, the frozen multi-step framework tracks the global shape and phase changes flawlessly because its path is governed strictly by the full-rank deterministic seasonal matrix (X) rather than short-term autoregressive noise.

3.4.2 Accuracy Evaluation

The quantitative accuracy assessment consolidated in Figure 9 provides definitive statistical proof regarding the predictive reliability of both forecasting paradigms.

	Model	MAE	Correlation
0	OLD Multi-step (24h)	4411.82 MW	0.9789
1	NEW 1-step Rolling	4631.97 MW	0.5410

Figure 9: Results of accuracy testing using different statistical measures

The empirical indices detailed in Figure 9 reveal that the 24-hour ahead frozen multi-step architecture delivers superior global tracking accuracy. It minimizes the Mean Absolute Error (MAE) to **4411.82 MW**, outperforming the paper’s rolling window approach which yields an inflated MAE of 4631.97 MW. Furthermore, the multi-step framework achieves an exceptional linear correlation coefficient of **0.9789** against the true load series, indicating a highly reliable fit. In contrast, the rolling window model experiences a severe degradation in its directional alignment, collapsing to a correlation baseline of 0.5410. Inferences from these comparative tests imply that for large-scale utility networks embedded with rigid socio-economic schedules, the explanatory power of global deterministic multi-seasonality heavily outweighs local autoregressive dependencies, confirming that the multi-step setup is capable of forecasting the complex consumption points of the power grid with a fairly high degree of structural security.

3.5 Limitations and Future Directions

Despite the highly informative and robust empirical results generated by the proposed framework, several limitations warrant structural consideration. First, the severe departure from normality observed in the ARIMAX residuals, as quantifiably verified by the extreme Kurtosis index of 14.58 and visually confirmed by the pronounced S-curve in the Q-Q plot, suggests that additional non-cyclical latent factors influencing electricity load movements are not fully captured by the current model specifications. Because the exogenous matrix relies purely on deterministic temporal dummy indicators (hour, day, month), the model lacks real-time environmental context to adapt flexibly to sudden climate anomalies. Future research could explore alternative specifications by directly integrating continuous high-frequency weather time-series (such as real-time temperature fluctuations, humidity indices, and wind speed) into the exogenous regressor space (ARIMAX) to eliminate the amplitude underprediction encountered during unexpected extreme heatwaves.

Moreover, despite the fact that the secondary GARCH(1, 1) model successfully addresses and isolates the system’s volatility clustering, it operates under the strict assumption that the conditional variance follows a symmetric functional form. In large-scale power grid infrastructures, the variance dynamics triggered by an extreme load surge (e.g., cooling emergency shocks during heatwaves) often exhibit fundamentally different grid stress characteristics compared to an unexpected load drop. Future studies can investigate the efficacy of different asymmetric volatility models, such as Exponential GARCH (EGARCH) or Glosten-Jagannathan-Runkle GARCH (GJR-GARCH), which possess the unique mathematical capacity to better capture leverage effects, directional variance feedback, and structural asymmetries in highly volatile energy utility workloads.

4. Conclusion

This study developed and evaluated a two-stage hybrid ARIMAX(2, 1, 5)-GARCH(1, 1) framework for forecasting hourly electricity consumption in the PJM East Region (PJME), successfully addressing the core challenges of multiple seasonality and conditional heteroskedasticity. By deploying a global estimation paradigm across a large-scale sample space of 145,366 observations, the linear ARIMAX layer successfully offloaded complex overlapping daily and weekly cycles through a full-rank deterministic exogenous dummy matrix, eliminating the risk of parameter dimensionality explosion. Concurrently, the non-linear GARCH layer successfully structuralized the volatile residual dynamics, proving a high sensitivity to immediate shocks ($\alpha_1 = 0.7040$) alongside rapid volatility dissipation mechanics ($\beta_1 = 0.0568$) unique to power grid infrastructures.

A major contribution of this paper is the rigorous comparative benchmarking between a 24-hour-ahead

frozen multi-step horizon and a 1-step-ahead adaptive rolling window strategy. The experimental results overturned conventional time-series assumptions:

- **The Rolling Approach Constraint:** Due to severe short-term autoregressive memory inertia, the 1-step rolling approach encountered systemic phase lags during sudden load transitions, yielding a higher MAE of 4631.97 MW and a severely degraded correlation of 0.5410.
- **The Multi-step Approach Resilience:** Conversely, the frozen multi-step framework demonstrated powerful global tracking resilience, minimizing the MAE to **4411.82 MW** and maintaining an exceptional correlation coefficient of **0.9789** with actual demand.

These findings empirically prove that for large-scale utility grids embedded with rigid social schedules, the explanatory power of global deterministic multi-seasonality heavily outweighs local autoregressive dependencies.

From a practical power systems engineering perspective, the empirical insights generated by this framework provide critical operational utility:

- **Day-Ahead Power Dispatch:** The superior global path alignment of the frozen multi-step structure provides a fundamentally more reliable and statistically secure framework for day-ahead electricity market scheduling and unit commitment under uncertainty.
- **Grid Safety and Reserve Planning:** While the rolling framework remains valuable for localized, real-time hourly balancing adjustments, the time-varying variance envelopes generated by the secondary GARCH block offer dispatchers a dynamic mathematical tool to accurately calculate spinning reserve margins during volatile climate shocks.

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